

US Department of Agriculture (USDA)



Statement of Work

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Project

Purchase of Li-Cor Gas Analyzer Components

General Information

1.0 Scope of Work

Provide a brief statement of the overall project, including goals and objectives.

Purchase Li-Cor 8250 Extension Manifold and (4) additional Chamber Cable and Tubing Bundles (9982-009).

2.0 Background

Provide a general description of the requirement, describing the acquisition history (if applicable), the current environment, and the vision of the future of the requirement. If information technology is a factor, provide pertinent technical facts.

The National Laboratory for Agriculture and the Environment has been using Li-Cor 8250 products extensively since 2020. The 8250 Extension manifold connects via a daisy chain method to other 8250's to expand the number of gas analyzer chambers collecting data. The cable and tubing are specifically designed to work with the LI-8250 multiplexer using Li-Cor cable-and-tubing assemblies.

Equipment Requirements

3.0 Technical Requirements / Tasks

Li-Cor 8250-01 Extension Manifold

- **Sample Volume:** 27.9 cm³
- **Weatherproof Rating:** Tested to IEC IP55 standard
- **Operating Range**
 - **Temperature:** -20 to 45°C
 - **Humidity:** 0 to 95% RH, non-condensing
- **Plumbing**
 - **Flow rate to/from chambers:** ~2 to 3 lpm
 - **Pump type:** Diaphragm
- **Communication**
 - **LED Indicators:** Power, Ready
 - **Input Port Connector:** Full-duplex RS-422 communication, 115,200 baud, plus 24 VDC power
 - **Output Port Connector:** Full-duplex RS-422 communication, 115,200 baud, plus 24 VDC. Each output is current-limited to ~1.1A
 - Combined steady state power consumption for all chambers should not exceed 1A.
- **Power**
 - **Power Requirements:** Input voltage range is supplied by Li-8250 Multiplexer. Typical 24V and 1.8A max
 - **Alternate Power Input:** 21.9 to 25.5 VDC 80 W

Li-Cor Chamber Cable and Tubing Bundle (9982-056)

- **Length:** 15 meters
- **Total Tube Length:** 30 meters
- **Total Tube Volume:** 237.5 cm³
- **Cable Connectors:** Sealed power and data cables for 8200-104/C Long-Term Chambers

4.0 Government Furnished

N/A

5.0 Deliverables / Schedule

N/A

6.0 Travel

N/A

7.0 Contractor's Key Personnel

Li-Cor authorized distributor.

Li-Cor

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8.0 Security Requirements

N/A

9.0 Data Rights

N/A

10.0 Section 508 – Electronic and Information Technology Standards

N/A

Attachment